Rainfall–Credit Forecasting Model: Design & Documentation Plan

Term paper project, Agricultural Economics. Scope: Bihar, Maharashtra and All-India. Reference paper: De & Vij (2013), "Are Banks Responsive to Exogenous Shocks to Credit Demand in Rural Economies? District-level Evidence from India."

1. Project Purpose and Objectives

Research question. Does the cyclicality of formal agricultural credit differ across weather, production and price shocks in Maharashtra and Bihar, and what does that imply for credit under different drought and rainfall scenarios?

Model objective. Build a forecasting and scenario tool that estimates how formal agricultural credit (bank credit outstanding to agriculture) is expected to change for a given level of rainfall deficit or excess, separately for Bihar, Maharashtra and All-India.

Specific objectives

1. Estimate the historical response of agricultural credit to rainfall shocks (drought and excess rain), yield shocks and price shocks.
2. Test whether credit is countercyclical (rises in distress) or procyclical (falls in distress) in each region.
3. Convert estimates into a scenario simulator: choose region, rainfall departure or drought category, and see the predicted change in credit with an uncertainty band.
4. Validate the model out of sample and be transparent about its limits.
5. Deliver a term paper plus a small interactive app as the "model" component.

2. What the Pilot Paper Gives Us (and What It Does Not)

Key design ideas to reuse

Idea in De & Vij (2013)	How we use it
Drought defined two ways: SPI below -1, and rainfall below 75% of long-period average (LPA)	Use rainfall departure from LPA as the main shock variable; add SPI as robustness if available
Timing: monsoon in calendar year t-1 affects credit at end of fiscal year t (March)	Align every shock to the preceding monsoon (June to September) before matching to end-March credit

Idea in De & Vij (2013)	How we use it
Log of agricultural credit as the outcome, with district and year fixed effects	Log credit (or log growth) with state fixed effects and year controls
Control for other bank credit (proxy for banking development)	Include log non-agricultural credit or state GDP as a control
Falsification test using neighbouring districts	Adapt: placebo test using a state with no shock that year, or lead (future) rainfall as a placebo
Dynamics: event-study around drought year	Replicate as lags 0 to 3 years of rainfall shock
Result: credit rises about 3 to 5% after a drought; the rise is in average loan size, not account numbers; private banks respond more than public banks	Use as a prior and as a benchmark to compare against our state-level estimates

Important differences from our project

- The paper uses district-level data (up to 640 districts, 1993 to 2010). Our plan uses **state-year** data. This means far fewer observations and less statistical power, so estimates will be noisier and the model must be kept simple.
- The paper studies drought as a demand shock to credit. Our question adds **production and price shocks** and asks about cyclicality, which is broader. We should state clearly that our model describes association and conditional prediction, not clean causal identification, unless we explicitly use the rainfall-only specification.
- The paper does not forecast. Forecasting is our extension, so we need an out-of-sample validation section the paper does not have.

3. Data Plan

3.1 Data catalogue

Variable	Source	Frequency and coverage	Notes and cautions
Agricultural credit outstanding (state-wise)	RBI, *Handbook of Statistics on Indian States*, Table "State-wise Credit to Agriculture by Scheduled Commercial Banks" (data from Basic Statistical Returns); also available through RBI's DBIE portal	Amount outstanding at end-March; series runs from 1990 into the 2020s	Older tables are in ₹ billion, newer ones in ₹ crore; some editions round to the nearest 100 crore. Use the most recent edition with unrounded values and splice carefully
Rainfall	IMD sub-division-wise annual rainfall (published in the RBI Handbook, sourced from IMD's Annual Climate Summary); IMD Pune and the IMD data supply portal for monthly, seasonal and district data	Annual by meteorological sub-division	Bihar is one sub-division. Maharashtra has four (Konkan and Goa, Madhya Maharashtra, Marathwada, Vidarbha), so a state rainfall index must be built by area-weighting the sub-divisions
Rainfall normals (LPA) and monsoon departures	IMD state and sub-division monsoon reports	Seasonal (June to September)	Use the monsoon season, not the calendar year, for the shock

Variable	Source	Frequency and coverage	Notes and cautions
Crop area, production, yield	Ministry of Agriculture, *Agricultural Statistics at a Glance*; state directorates of economics and statistics (Bihar e-Statistics portal, Maharashtra agriculture department); the RBI Handbook also reproduces state-wise yield and production tables	Annual, by crop and state	Use total foodgrains yield plus the major crops for each state (see below). Recent years are sometimes provisional or revised
Crop prices	Agmarknet (wholesale mandi prices), Minimum Support Prices from the Commission for Agricultural Costs and Prices (CACP), Office of the Economic Adviser WPI (agricultural commodities)	Daily or monthly (Agmarknet), annual (MSP)	Aggregate to the crop year; deflate with a general price index
Deflator and macro controls	State GSDP and deflator (MoSPI, RBI Handbook), CPI	Annual	Needed to convert nominal credit to real terms
Policy events	Farm loan waivers (national 2008; state schemes including Maharashtra 2017), interest subvention and Kisan Credit Card changes, RBI natural-calamity guidelines	Event dates	Code as dummy variables; the paper found RBI circular changes had little effect, so test but do not over-weight

Variable	Source	Frequency and coverage	Notes and cautions
All-India series	RBI Handbook all-India agricultural credit; IMD all-India monsoon departure; Agricultural Statistics at a Glance all-India yield	Annual	Use as a third region in the same specification

All sources above are public. Before finalising, confirm table numbers and latest available year, because RBI renumbers tables across editions.

3.2 Crop selection (price and production shocks)

- **Bihar:** rice, wheat, maize, pulses, and possibly sugarcane and potato. Foodgrains dominate.
- **Maharashtra:** cotton, sugarcane, soybean, jowar, tur/pulses, onion. More cash-crop heavy, so price exposure is higher.
- **All-India:** foodgrains, oilseeds and sugarcane aggregates.

Build one crop-weighted price index per state using area or production shares in a base period, so the index does not shift just because the crop mix changed.

3.3 Known data problems and how we handle them

1. **Stock, not flow.** Credit is amount outstanding, so it moves slowly and includes rolled-over loans. Model log growth as well as log level, and discuss that a rise can reflect restructuring, not only new lending (the paper tests this indirectly using personal loans; we cannot, so state it as a limitation).
2. **Location of lending.** State-wise credit by scheduled commercial banks may be recorded by the location of the branch. Metro centres (Mumbai in Maharashtra, Delhi) can show large corporate or indirect agricultural lending that does not reach farmers. Check whether a direct-finance series is available for Maharashtra; if not, run a robustness check excluding Mumbai-heavy years or use a farm-credit proxy and note the bias.
3. **State boundary change.** Jharkhand was carved out of Bihar in 2000. Pre-2000 Bihar figures include Jharkhand. Either use a post-2000 sample, or construct a combined "undivided Bihar" series for the whole period.
4. **Structural breaks.** Priority-sector lending rules, the 2008 waiver, bank mergers and the Kisan Credit Card expansion. Include period dummies and check stability.

5. **Nominal to real.** Deflate by a price index before any elasticity statement.
6. **Small sample.** Roughly 25 to 30 annual observations per region. Limit regressors, avoid overfitting, and report uncertainty honestly.

4. Variable Construction

Outcome (Y): log real agricultural credit outstanding at end of fiscal year t, and its year-on-year growth.

Weather shock (rainfall, from the monsoon of calendar year t-1):

- Monsoon rainfall departure from LPA in per cent.
- Drought categories (scenario levels), using IMD's standard departure bands (confirm the exact cut-offs in IMD documentation): normal (within about ±19%), deficient (about -20% to -59%), and scanty/severe (-60% or worse). Add a moderate-drought flag at below 75% of LPA, as in the paper.
- Excess rainfall flag (for example above +20%) to capture flood-type shocks.
- Optional: SPI below -1 as a robustness drought flag.

Production shock: deviation of log yield from its state-specific trend (detrended residual), or a simple negative yield-shock dummy for years more than one standard deviation below trend.

Price shock: deviation of the real crop price index from trend, and a dummy for sharp price falls (revenue risk for farmers).

Controls: log non-agricultural bank credit or real GSDP, policy dummies (waiver years, scheme changes), and a time trend.

5. Modelling Approach

5.1 Stage 1: Cyclicality (the research question)

Panel regression on Bihar and Maharashtra, following the paper's equation 1:

$\log(\text{Credit})_{st} = \alpha_s + \nu_t$ (or trend) $+ \delta_1 \cdot \text{Rainfall shock}_{s,t-1} + \delta_2 \cdot \text{Yield shock}_{s,t-1} + \delta_3 \cdot \text{Price shock}_{s,t-1} + \beta \cdot \text{Controls}_{st} + \varepsilon_{st}$

with state fixed effects. Then allow different slopes by state (interaction with a Maharashtra dummy) to test whether cyclicality differs. A positive δ means countercyclical credit (credit rises in distress); a negative δ means procyclical (banks pull back).

Use robust or clustered standard errors. With only two states, clustering by state is unreliable, so prefer heteroskedasticity- and autocorrelation-robust (Newey-West) errors and report wild-bootstrap p-values as a check.

5.2 Stage 2: Forecasting model

Start simple and add complexity only if it improves out-of-sample accuracy.

Model	Role
Naive (credit grows at trend)	Baseline to beat
Autoregressive distributed lag (ARDL) of credit growth on rainfall departure (lags 0 to 2), yield and price shocks	Main model, separate for Bihar, Maharashtra and All-India
Panel fixed-effects with state-specific slopes	Pools Bihar and Maharashtra to stabilise estimates
Regularised regression (ridge or lasso) or gradient boosting	Optional comparison; with so few observations treat as a robustness check, not the headline model

The model must respect the timing rule: only information available before the credit observation can be used to forecast it.

5.3 Stage 3: Scenario engine

Input: region, rainfall departure (slider or drought category), optional yield and price shock, forecast horizon of 1 to 3 years. Process: apply fitted coefficients to the chosen shock, add the baseline trend, and produce a central forecast. Output: predicted change in credit (% and ₹ crore), a prediction interval from residual bootstrap, and a comparison with the baseline (no shock) path.

5.4 Validation

- **Rolling-origin (expanding window) cross-validation:** train on years up to T, forecast T+1, repeat. Report MAE, RMSE and MAPE against the naive baseline.
- **Leave-one-drought-out check:** hold out a known drought year (for example 2002, 2009, 2014 or 2015 in Maharashtra; 2009 and 2010 in Bihar) and see if the model predicts the credit response.
- **Placebo tests:** use future rainfall (t+1) to predict credit at t; it should have no effect.
- **Robustness:** alternative drought thresholds (70%, 75%, 80% of LPA), SPI-based flag, excluding waiver years, post-2000 sample only.

- **Compare with the paper's benchmark:** a drought-year rise of roughly 3 to 5% in district data. If our state-level estimate is wildly different, investigate before reporting.

6. System Logic (Pipeline)

1. **Ingest:** download tables (CSV or Excel) from RBI, IMD, Agricultural Statistics and price sources into a `/data/raw` folder; never edit raw files.
2. **Clean:** standardise state names, units (₹ crore), fiscal vs calendar years, and splice series across table editions; log every manual fix in a data-cleaning log.
3. **Align timing:** map monsoon year t-1 to credit at end-March of fiscal year t.
4. **Construct variables:** rainfall departure, drought flags, detrended yield and price shocks, deflated credit.
5. **Estimate:** Stage 1 cyclicality regressions; Stage 2 forecasting models.
6. **Validate:** cross-validation, placebo and robustness tables.
7. **Export:** save coefficients, residuals and forecasts as files the app reads.
8. **Serve:** interactive app reads the saved outputs (no model re-fit on each click).
9. **Document:** methodology, data dictionary, and a reproducibility note.

7. Interface (UI) and User Experience (UX) Plan

Users: the course instructor and classmates (reading results quickly), and you (running scenarios while writing). Design for a phone screen first, then desktop.

Principle: a scenario in three taps, with a plain-language answer first and technical detail on demand.

Screens

Screen	Content
1. Scenario simulator (home)	Region selector (Bihar, Maharashtra, All-India); rainfall control as a slider from -60% to +40% of normal with labelled bands (Normal, Deficient, Severe drought, Excess); optional toggles for yield and price shocks; forecast horizon selector.

Screen	Content
2. Result card	One headline sentence, for example "A 30% monsoon deficit in Maharashtra is associated with about X% higher/lower agricultural credit next year (range A to B)." Below it, a fan chart showing baseline vs scenario credit path with the uncertainty band, and a small table in ₹ crore.
3. Compare regions	The same shock applied to Bihar, Maharashtra and All-India side by side, showing who is countercyclical and who is procyclical.
4. History explorer	Time series of rainfall departure, yield, price index and credit with drought years shaded, so users see the raw data behind the model.
5. Method and data	Plain-language explanation, model equation, validation scores, data sources, and a limitations box.

UX rules

- Always show uncertainty (intervals), never a single bare number.
- Show a warning when the chosen scenario lies outside the range seen in history (for example rainfall deficit worse than any observed year in that state).
- Use consistent colours: one colour per region; red for drought, blue for excess rain.
- Label the unit (₹ crore, real or nominal) on every chart.
- Keep text short; put technical detail behind expandable sections.

8. Tool Stack

Need	Recommended tool
Data cleaning and modelling	Python (pandas, numpy, statsmodels, linearmodels for panel regressions, scikit-learn for validation) or R (fixest, plm, forecast)
Time-series diagnostics	statsmodels (ADF/KPSS unit-root tests, Breusch-Godfrey, Newey-West errors)
Visualisation	matplotlib/plotly for the paper; plotly or Chart.js for the app

Need	Recommended tool
Interactive app	Option A: a single self-contained HTML page that embeds the saved coefficients (easy to share, works on mobile). Option B: Streamlit app (more flexible, needs hosting). Start with A unless you need heavier computation
Version control and reproducibility	Git repository, a `README`, a fixed `requirements.txt`, and a script that rebuilds every table and figure
Writing	A document editor with citations; a reference manager such as Zotero
Data wrangling for PDF tables	Tabula or camelot for extracting tables from RBI PDF files, followed by manual checking

9. Documentation Deliverables

1. **Term paper** (main deliverable): introduction, literature, institutional background, data, empirical strategy, results, forecasting and validation, policy implications, limitations.
2. **Data dictionary:** every variable, definition, unit, source, and transformation.
3. **Model card:** what the model does, what it cannot do, training period, performance numbers, and known failure cases.
4. **Reproducibility package:** code, raw data links, cleaning log, and run instructions.
5. **App user guide:** one page with screenshots and example scenarios.

10. Suggested Term Paper Outline

1. Introduction and research question
2. Related literature (rainfall and farm output, rural credit, bank behaviour in shocks; the pilot paper)
3. Institutional background: priority-sector lending, RBI calamity guidelines, loan waivers, Bihar and Maharashtra agriculture
4. Data and variable construction
5. Empirical strategy (Stage 1 and Stage 2)
6. Results: cyclicality by shock type and by state
7. Forecasting results and scenario analysis
8. Robustness and validation

9. Discussion, policy implications, limitations
10. Conclusion

11. Work Plan

Phase	Tasks	Output
1. Setup	Finalise question, list variables, set up folders and repository	Project skeleton
2. Data collection	Download RBI credit tables, IMD rainfall, yield and price data; extract PDF tables	Raw data folder
3. Cleaning and construction	Align timing, deflate, build shock variables, document fixes	Clean state-year panel
4. Stage 1 estimation	Cyclicality regressions, state comparison, event-study lags	Results tables
5. Stage 2 forecasting	ARDL and panel models, cross-validation, scenario engine	Forecast files
6. App build	Build simulator and compare screens from saved outputs	Working prototype
7. Writing and checks	Paper draft, robustness tables, model card, proofreading	Final submission

12. Risks and Limitations to State Openly

- **Small sample and low power:** state-year data give few observations; a null result may reflect weak power rather than no effect.
- **Association, not full causality:** rainfall is plausibly exogenous, but yield and price shocks are partly endogenous to farmer and policy responses.
- **Credit measurement:** outstanding stock mixes new lending, rollovers and accrued interest; metro-booked lending may distort Maharashtra.
- **Policy confounding:** loan waivers and scheme changes coincide with some droughts.
- **Extrapolation:** scenarios beyond historical range are unreliable; the app should flag them.
- **Climate non-stationarity:** past rainfall-credit relationships may not hold under a changing monsoon pattern.

13. Immediate Next Steps

1. Confirm with your instructor whether state-year scope and a forecasting component are acceptable, and whether district data would be expected.
2. Download the RBI state-wise agricultural credit table and the IMD sub-division rainfall table first; these two files decide whether the project is feasible and take the longest to clean.
3. Decide the sample period (full period with a Jharkhand adjustment, or post-2000 only).
4. Build the first panel and run a one-line regression of credit growth on lagged rainfall departure for each state; this quick check tells you whether the signal is strong enough to forecast.